



HỒ MINH DŨNG

Robotics & Mechatronics Engineer

Contact

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Skills

ROBOTICS & AUTOMATION

- ROS2 & Nav2
- SLAM & Path Planning
- Computer Vision
- Sensor Fusion

EMBEDDED SYSTEMS & IOT

- ESP32, STM32, RK3399
- CAN, RS485, MQTT, LoRa
- IoT Integration
- OTA Updates

PROGRAMMING & TOOLS

- C/C++, Python
- Linux, Docker
- Git, ESP-IDF

LANGUAGES

- Bachelor's degree instructed in English
- English – IELTS 6.5 (2019)
- Japanese – JLPT N5

Summary

Robotics Engineer specializing in autonomous navigation, embedded IoT, and computer vision. Experienced in developing ROS2-based service robots for healthcare, events, and agriculture, working across the full stack from embedded firmware to navigation algorithms. Aspire to develop reliable autonomous robot solutions for complex real-world environments.

Education

RMIT University Vietnam

Jul 2019 - Apr 2024

Bachelor of Engineering - Robotics & Mechatronics

Honours: Second Class Division A (2A) | GPA: 3.1/4.0

Capstone Project: Factors Affecting Robot Accuracy in Robot-Vision Applications and Solutions

ABB Automation and Electrification (Vietnam)

- Researched industrial robot arm and vision system principles
- Analyzed factors affecting vision-guided robot accuracy

Experience

Robotics and IoT Engineer

Dec 2024 - Present

Phenikaa-X JSC / Phenikaa NEO

Medication Delivery Robot

- Designed IoT control board with Ethernet, WebSocket, 4G SIM integration (C/ESP32)
- Created ESP-Now protocol for inter-robot collision avoidance
- Built IoT speaker system with LoRa communication, web server for configuration, and OTA updates

Assistance/Reception Robot

- Developed motor control (CAN, RS485, VESC) on RK3399/Aidlux
- Integrated facial recognition for tracking and QR code recognition for event check-in
- Designed real-time WebSocket interface for sensor telemetry and IoT device control
- Developed elevator control system: WT32-ETH01 for cabin control, Python server for cloud commands

Robotics Engineering Intern

Jul 2024 - Nov 2024

Phenikaa-X JSC / Phenikaa NEO

Outdoor Autonomous Robot

- Developed navigation system using ROS2 and Nav2 for outdoor environments
- Built odometry and control algorithm for Ackermann model
- Integrated GPS/RTK and path planning with reverse motion

Embedded Software Intern

Nov 2023 - May 2024

Bosch Global Software Technologies Vietnam

- Tested AUTOSAR projects focusing on UDS services over CAN protocol
- Developed DoIP testing library and optimized FBL testing tool
- Created automated LDF parsing and LIN test generation tools